MINI PROJECT

Title

ResearchSphere AI: An Intelligent Collaborative Research Assistant using LLM, RAG and Knowledge Graph

Objective

The mini project aims to develop an AI-powered collaborative research platform that assists students and researchers throughout the initial stages of research. Unlike conventional AI chatbots, the system provides a structured research workspace where users can upload research papers, understand complex concepts through AI, visualize knowledge using a Knowledge Graph, collaborate with guides and peers, receive academic feedback, and prepare presentations and viva sessions. The integration of Retrieval-Augmented Generation (RAG) ensures responses are grounded in uploaded documents, while the Knowledge Graph helps users understand relationships between concepts instead of reading lengthy papers manually.

USER ROLES

Student / Researcher

The Student or Researcher is the primary user of the platform. They can create projects, upload papers, interact with AI, participate in discussions, collaborate with team members, receive guide feedback, and monitor their progress.

Guide / Supervisor

Each guide supervises three to four students. Guides review uploaded work, provide comments, assign marks, approve milestones, monitor progress, and interact with students through collaboration rooms.

Administrator

The administrator manages users, assigns guides, creates research groups, monitors platform usage, and maintains system configurations.

MINI PROJECT MODULES

Module 1 – User Authentication & Profile Management

Users register securely using JWT authentication and maintain academic profiles with research interests, skills, institution details, and specialization. These profiles later support collaborator recommendations and research analytics.

Module 2 – Research Workspace

Each user has a dedicated research workspace to upload, organize, preview, search, categorize, and manage research papers, datasets, presentations, and notes. The workspace serves as the central repository for all research activities.

Module 3 – Document Processing Engine

When a document is uploaded, the system automatically extracts text, cleans unnecessary formatting, identifies sections such as abstract and methodology, and prepares the content for embedding generation and knowledge extraction.

Module 4 – Knowledge Graph Construction ⭐

This is one of the core modules.

Instead of storing only text, the system extracts entities such as algorithms, datasets, programming languages, diseases, organizations, authors, frameworks, and technologies. It then identifies relationships among them and stores them in Neo4j.

Example:

CNN
  |
uses
  |

MRI Dataset
  |
belongs to
  |
Medical Imaging

The Knowledge Graph enables semantic understanding rather than keyword searching.

Module 5 – Knowledge Graph Visualization ⭐

Users can explore uploaded research visually instead of reading hundreds of pages.

The graph allows users to

- Expand related concepts
- Explore connected technologies
- Identify frequently occurring methods
- Navigate research topics
- Understand relationships among concepts

This module distinguishes the project from ordinary AI chat applications.

Module 6 – AI Research Assistant (LLM + RAG + KG)

This module combines three technologies:

- LLM
- Retrieval-Augmented Generation
- Knowledge Graph

Workflow:

Question

↓

Retrieve relevant document chunks

↓

Retrieve related entities from Knowledge Graph

↓

Merge both contexts

↓

LLM generates answer

This produces richer and more reliable answers than using only an LLM.

Module 7 – AI Explain Mode

The AI automatically adjusts explanations according to the user's knowledge level.

Levels include:

- Beginner
- Undergraduate
- Postgraduate
- Researcher

The same concept is explained differently depending on the selected audience.

Module 8 – AI Summary Generator

Generates

- Abstract
- Key findings
- Methodology summary
- Literature summary
- Conclusion summary

Users can choose between short, medium, or detailed summaries.

Module 9 – AI Citation Generator

Automatically generates citations in APA, IEEE, MLA, Chicago, and Harvard formats while identifying missing references.

Module 10 – AI Presentation Assistant

Creates presentation-ready content including:

- Introduction
- Problem Statement
- Objectives
- Literature Review
- Methodology
- Results
- Future Scope

Module 11 – AI Viva Preparation

Generates viva questions categorized as:

- Easy
- Medium
- Advanced

Provides model answers based on the uploaded research.

Module 12 – Research Community

Researchers can

- Share research ideas
- Post papers
- Ask questions
- Comment
- Follow researchers
- Like discussions

Module 13 – Collaboration Room

Each guide supervises a private collaboration room where assigned students share progress, discuss ideas, upload weekly reports, and receive announcements.

Module 14 – Guide Review System

Whenever a student uploads work, the guide can

- Review documents

- Add annotations
- Suggest improvements
- Assign marks
- Approve submissions
- Request resubmission

Students receive instant notifications.

Module 15 – Progress Tracking Dashboard

The dashboard visualizes

- Upload history
- AI usage
- Research progress
- Guide feedback
- Pending tasks
- Collaboration activity
- Milestone completion

MAIN PROJECT

The main project extends the mini project into a **complete AI-powered Research Management and Publication Platform**.

Additional Roles

- Reviewer
- Editor
- Panel Member
- Journal Coordinator
- Department Head
- External Researcher
- Institutional Administrator

MAIN PROJECT MODULES

Module 16 – AI Research Gap Suggestion

Instead of claiming to "discover" research gaps, the system analyzes uploaded papers and suggests **potential underexplored areas** by comparing topics, methods, and keywords across documents. Researchers can use these suggestions as starting points for future work.

Module 17 – Global Knowledge Graph ⭐⭐⭐⭐⭐

The Knowledge Graph expands beyond individual documents and connects every uploaded research paper.

It links

- Authors
- Universities
- Algorithms
- Research Domains
- Datasets
- Publications
- Journals
- Projects

creating an institution-wide research network.

Module 18 – Reviewer Management System

Once a guide approves a submission, the paper is forwarded to reviewers.

Reviewers can

- Read submissions
- View AI-generated summaries
- Inspect the Knowledge Graph
- Add comments
- Recommend acceptance or revision
- Submit structured review reports

Module 19 – AI Reviewer Assistant

Before human review, AI analyzes the manuscript for:

- Grammar
- Structure
- Citation consistency
- Readability
- Methodology clarity
- Missing references
- Possible duplicate content (if integrated with similarity tools)

It generates a preliminary review report to assist reviewers.

Module 20 – Editorial Management

Editors oversee the complete review workflow by assigning reviewers, tracking review progress, resolving conflicts between reviews, and making publication decisions.

Module 21 – Panel Evaluation System

Panel members evaluate student projects using structured rubrics.

Evaluation includes:

- Innovation
- Technical implementation
- Documentation
- Presentation
- Demonstration
- Research contribution

Marks are automatically combined into a final score and ranking.

Module 22 – Journal Submission Workflow

Research follows a complete lifecycle:

Student

↓

Guide Approval

↓

Reviewer Evaluation

↓

Editor Decision

↓

Publication Repository

Module 23 – Research Repository

Stores:

- Research papers
- Project reports
- Source code
- Datasets
- Presentations
- Publications

Supports advanced semantic search.

Module 24 – AI Collaborator Recommendation

Using profiles, uploaded work, and the Knowledge Graph, the system recommends researchers with similar interests and complementary expertise for collaboration.

Module 25 – AI Literature Explorer

Users can explore related papers through an interactive graph instead of traditional keyword searches, helping them discover connections among topics and methodologies.

Module 26 – AI Research Analytics

Provides metrics such as:

- Research activity
- Collaboration frequency
- Popular topics
- Guide workload
- Student progress
- Publication trends

Module 27 – AI Meeting Assistant

After guide–student meetings, AI generates:

- Meeting summary
- Key discussion points
- Action items
- Deadlines
- Progress updates

Module 28 – Smart Deadline & Notification Manager

Tracks:

- Synopsis submission
- Weekly reviews
- Mid-term evaluation
- Final report
- Presentation schedule
- Publication deadlines

Module 29 – AI Research Recommendation Engine

Recommends:

- Relevant research papers
- Suitable journals
- Conferences
- Books
- Online courses
- Research tools

based on the user's interests and current project.

Module 30 – Institutional Research Dashboard

Provides administrators and department heads with insights into:

- Student progress
- Guide assignments
- Reviewer workload
- Publications
- Project rankings
- Department research output

Why this design is stronger

This structure gives your project a clear progression:

- **Mini Project:** A complete, working AI-assisted research collaboration platform focused on document understanding, guide supervision, collaboration, and Knowledge Graph visualization.
- **Main Project:** A full institutional research ecosystem that adds peer review, editorial workflows, publication management, analytics, and advanced AI assistance.

The Knowledge Graph is not just another feature—it becomes a **core intelligence layer** used by the AI assistant, collaborator recommendations, reviewer support, literature exploration, guide dashboards, and institutional analytics. This makes your project much more distinctive than a standard RAG chatbot while still being technically coherent and explainable during your viva.